D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

Mucosal melanoma.

February 2025 | DM 0225 | Q2 (DEEP Essay / Elaborate (FOCUS 2022–2026)) | **15 marks** | Target ~16 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

0. How to use this DEEP answer

Sections below are **high-yield, trial-dense content** written for a 15-mark Paper IV elaboration. After the deep core, additional revision modules (algorithms, supportive care, flash cards) extend the note to full booklet length.

EXAM PEARLS / HALLMARKS

- In the exam hall, prioritise the DEEP trial tables + NCCN 2026 boxes first.
- Draw at least one flowchart from the deep sections.

1. Introduction — why mucosal melanoma is a distinct entity

Mucosal melanoma (MM) arises from melanocytes in ectodermal mucosa — ~1–2% of melanomas in Western series but a higher fraction of melanoma in Asian/Indian cohorts where cutaneous UV-driven disease is less dominant. It is biologically and clinically distinct from cutaneous melanoma: different molecular drivers, later stage at diagnosis, richer local invasion, and inferior outcomes with surgery, radiotherapy, and checkpoint inhibitors.

Exam framing: sites → pathology/molecular (KIT/NRAS/BRAF) → staging work-up → local therapy (surgery/RT) → systemic CPI (± targeted) → prognosis vs cutaneous → NCCN cutaneous neoadjuvant contrast.

EXAM PEARLS / HALLMARKS

- Opening contrast sentence scores marks: ‘Unlike cutaneous melanoma, mucosal disease is UV-independent, KIT-enriched, and CPI-hyporesponsive.’

2. Epidemiology & anatomic sites

Major site groups: (1) Head and neck mucosal — sinonasal cavity, oral cavity, oropharynx, larynx (sinonasal often presents with epistaxis/obstruction). (2) Anorectal — rectal/anal melanomas; may mimic hemorrhoids. (3) Vulvovaginal — vaginal > vulvar mucosal primaries. Rarer: urinary tract, esophagus, gallbladder.

Age typically older adults; equal or site-dependent sex skew (vulvovaginal in women; anorectal slight female predominance in some series). No clear UV association — sun-protection counseling still relevant for cutaneous surveillance but not causal for MM.

- India/Asia: higher proportional burden of mucosal/acral vs superficial spreading cutaneous subtypes.
- Delayed diagnosis common → thicker/locally advanced primaries.
- Rule out metastatic cutaneous melanoma to mucosa (history, skin exam, regression scars).

Initial MM pathway

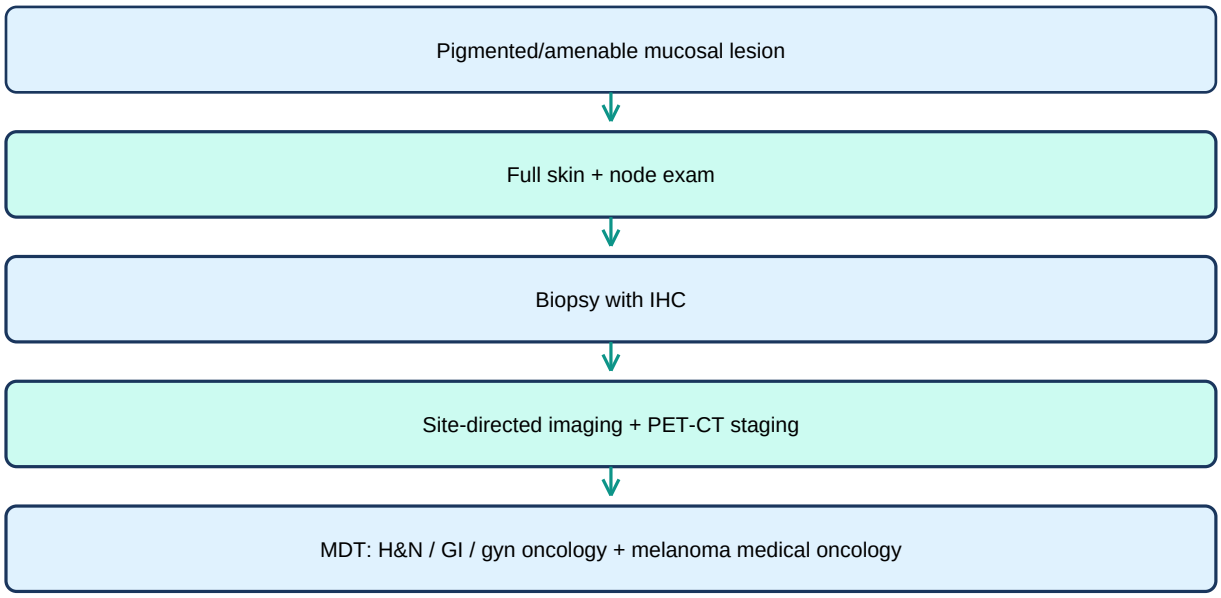


Figure: stepwise exam algorithm

3. Pathology & diagnosis

Histology: atypical melanocytes ± melanin; amelanotic tumors frequent — IHC mandatory (S100, SOX10, Melan-A/MART-1, HMB-45; cytokeratin negative helps exclude carcinoma). Breslow depth analogs and ulceration still reported when measurable; for sinonasal, local extent into bone/skull base often dominates staging language.

Differential: mucosal melanosis, carcinoma, sarcoma, lymphoma, clear cell sarcoma (EWSR1), metastatic cutaneous melanoma. Document margins meticulously — R0 rare in sinonasal disease.

- Always photograph and map lesions before wide resection.
- PD-L1 not reliable sole selector for CPI in MM.
- Send NGS panel covering BRAF, NRAS, KIT, NF1, SF3B1 (esp. uveal — different entity), TMB/MSI rarely high.

UV cutaneous	High TMB, BRAF V600 common
Mucosal	Low TMB, KIT/NRAS enriched
Uveal (contrast)	GNAQ/G11, BAP1 — not mucosal
IHC	SOX10/Melan-A confirm melanocytic
NGS	Actionable KIT / rare BRAF

Figure: mechanism / pathway map

4. Molecular biology — KIT, NRAS, BRAF (rarer)

KIT mutations/amplifications enriched in mucosal and acral melanomas (exon 11, 13, 17 hotspots). NRAS mutations occur at meaningful frequency. BRAF V600 mutations are substantially rarer than in cutaneous melanoma (~6–15% vs ~50%) — still test because MEK/BRAF inhibitors remain options if present.

Tumor mutational burden is typically lower → fewer neoantigens → partly explains inferior CPI response versus cutaneous melanoma. SF3B1 and other alterations appear in some mucosal series; fusions uncommon but

NGS may reveal trial options.

- Always sequence KIT before declaring ‘no targeted option’.
- KIT exon 11 often more imatinib-sensitive than exon 17 (pattern shared with GIST teaching).
- Do not start BRAF/MEKi without confirmed BRAF V600.

EXAM PEARLS / HALLMARKS

- Mnemonic: Mucosal → KIT/NRAS; Cutaneous → BRAF/UV-TMB; Uveal → GNAQ/G11.

5. Staging & risk assessment

AJCC cutaneous TNM is imperfect for MM; site-specific systems and clinical staging (local / regional / distant) guide practice. MRI for sinonasal/skull base and pelvic MM; endoscopic staging for anorectal; CT/PET for distant disease (lung, liver, bone).

Prognostic factors: site (anorectal often worse), tumor thickness/size, nodal status, resectability, LDH, performance status, and molecular findings. Five-year OS historically much lower than stage-matched cutaneous melanoma.

Trial	Key result / exam takeaway
Institutional MM series	Inferior OS vs cutaneous; late presentation
Meta-analyses CPI MM	ORR typically ~20–35% vs higher cutaneous

Table: landmark clinical trials

6. Local therapy — surgery

Surgery remains the backbone for localized disease when R0/R1-meaningful resection is feasible without catastrophic morbidity. Wide margins akin to cutaneous norms are often anatomically impossible in sinonasal/vaginal sites — balance oncologic clearance with function (vision, speech, continence, sexual function).

Elective node dissection is not uniformly standardized; sample clinically involved nodes. Sentinel node concepts less validated than cutaneous. Anorectal: abdominoperineal resection historically used; organ preservation debates exist for early lesions with RT/systemic combinations in specialized centers.

- Refer sinonasal MM to high-volume skull-base teams.
- Avoid piecemeal inadequate excisions that seed mucosa.
- Metastasectomy rarely curative but may palliate selected oligometastases after systemic control.

Local therapy decision

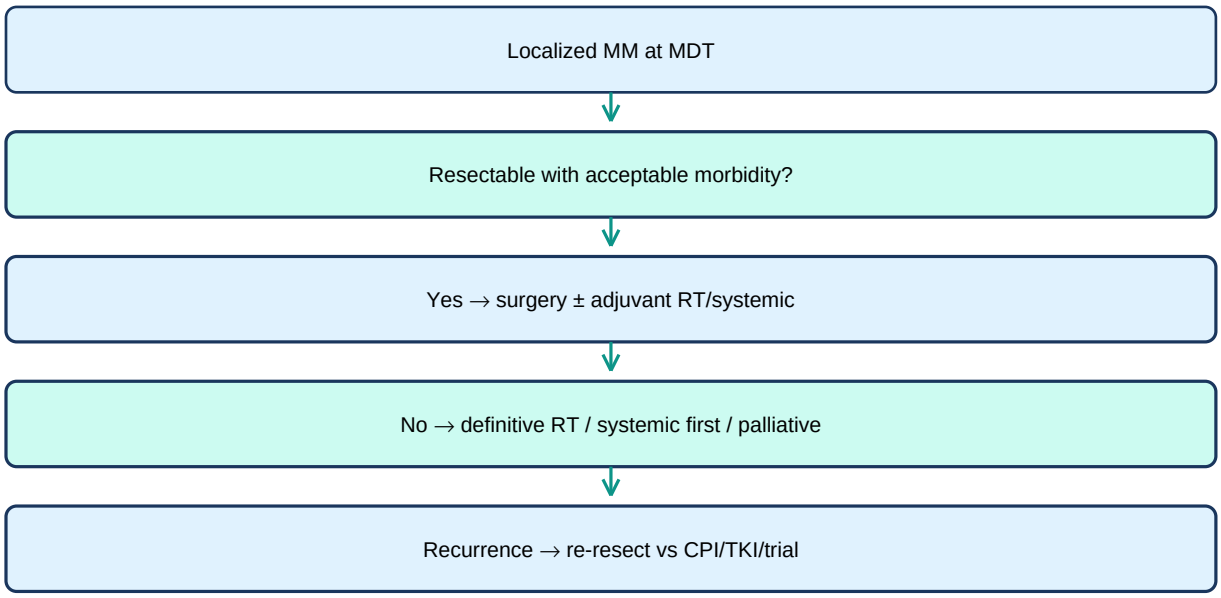


Figure: stepwise exam algorithm

7. Radiotherapy

Melanoma is relatively radioresistant, yet RT is important for local control in incompletely resected sinonasal/oral disease, vaginal primaries, and palliation of bleeding/pain. Techniques: IMRT/VMAT, stereotactic approaches for skull base; consider proton therapy discussion for sinonasal OAR sparing in selected centers.

Adjuvant RT after close/positive margins can improve local control without clear OS proof — state this nuance. Hypofractionation historically used in melanoma; modern conformal fractionation individualized. Combine carefully with CPI (recall additive dermatitis/mucositis and rare recall phenomena).

- Palliative RT for symptomatic metastases remains valuable.
- Document dental/endocrine/sexual late effects risk.

8. Systemic therapy — checkpoint inhibitors (CheckMate / KEYNOTE mucosal subgroups)

Anti–PD-1 monotherapy (nivolumab, pembrolizumab) and combinations (nivolumab+ipilimumab; nivolumab+relatlimab) are used by extrapolation from cutaneous paradigms, but mucosal subgroup analyses and dedicated series show lower ORR and shorter survival than cutaneous melanoma.

CheckMate 067 mucosal subgroups and pooled anti–PD-1 analyses: responses occur and durable benefit is possible — still offer CPI as standard systemic backbone for unresectable/metastatic MM absent a better targeted option. KEYNOTE-006/KEYNOTE mucosal cohorts similarly support pembrolizumab activity at a lower response rate.

Neoadjuvant CPI is transforming resectable cutaneous nodal melanoma (SWOG S1801, NADINA) — evidence in mucosal primaries is far thinner; do not uncritically copy cutaneous neoadjuvant algorithms without MDT caveats.

Trial	Key result / exam takeaway
CheckMate 067 (mucosal subgroup)	Nivo±ipi active but inferior outcomes vs cutaneous
KEYNOTE-006 / mucosal cohorts	Pembrolizumab activity; lower ORR than cutaneous
Pooled anti–PD-1 MM analyses	ORR ~23–30%; some durable remissions

RELATIVITY-047 context	Nivo+relatlimab — limited MM-specific level-1 data
SWOG S1801 / NADINA	Neoadjuvant CPI — cutaneous resectable nodal; contrast with MM

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Nivolumab	240 mg Q2W or 480 mg Q4W
Pembrolizumab	200 mg Q3W or 400 mg Q6W
Nivo + ipi (melanoma)	Nivo 1 mg/kg + ipi 3 mg/kg Q3W ×4 → nivo maintenance
Nivo + relatlimab	Fixed-dose combo per label Q4W

Table: doses and regimens

EXAM PEARLS / HALLMARKS

• Quote ‘lower ORR than cutaneous but still first systemic choice for most unresectable MM’.

9. Targeted therapy if KIT (and rare BRAF)

Imatinib: phase II data in KIT-mutant melanoma (including mucosal) show responses enriched for exon 11 mutations; median PFS modest — use when CPI contraindicated, after CPI, or in molecularly selected rapid progression. Dose often 400 mg daily (some escalate). Nilotinib/dasatinib studied with mixed results; ripretinib/other KIT inhibitors via trials.

BRAF V600–mutant MM (uncommon): dabrafenib+trametinib or encorafenib+binimetinib per cutaneous dosing. NRAS: MEK inhibitor trials historically; no robust standard — prefer CPI/trial. Avoid empiric KIT TKI without mutation.

Systemic algorithm for MM

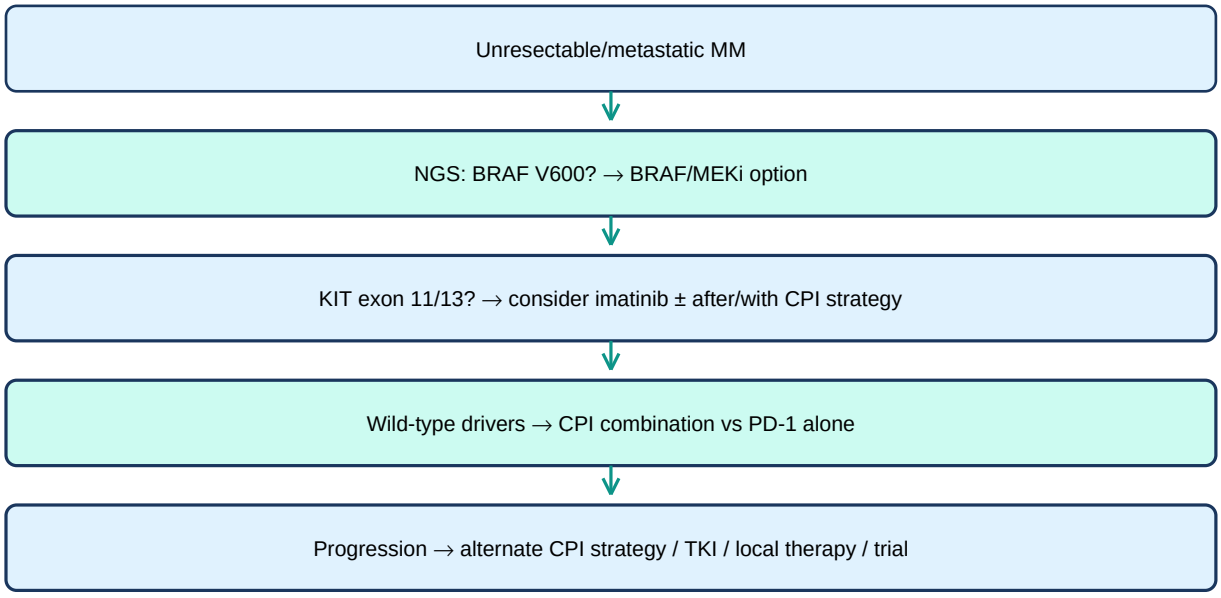


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Imatinib (KIT-mut MM)	400 mg PO daily (selected escalation)
Dabrafenib + trametinib	150 mg BID + 2 mg daily if BRAF V600
Encorafenib + binimetinib	450 mg daily + 45 mg BID if BRAF V600

Table: doses and regimens

10. Prognosis vs cutaneous melanoma

MM carries poorer prognosis stage-for-stage: delayed detection, difficult R0, early hematogenous spread (anorectal), and lower CPI sensitivity tied to low TMB. Five-year survival for advanced MM is substantially inferior to cutaneous cohorts in CPI landmark trials.

Counsel patients honestly while emphasizing that durable CPI responses still occur. Supportive care, nutrition, and psychosocial support are integral — facial disfigurement and sexual/continence morbidity need rehabilitation planning.

- Surveillance: frequent local exams + imaging; pattern-of-failure often local+distant.
- Second opinions at melanoma referral centers appropriate.

11. Multidisciplinary & site-specific pearls

Sinonasal: endoscopy, MRI, skull-base surgery, adjuvant RT, early CPI for unresectable. Oral: dental oncology. Anorectal: surgical oncology + RT + systemic; distinguish from squamous anal HPV disease. Vulvovaginal: gyn-oncology radicality vs function; RT for organ preservation discussions.

- Amelanotic anorectal MM — biopsy every atypical ‘hemorrhoid’.
- HPV testing does not diagnose melanoma — IHC does.
- Screen for synchronous cutaneous primaries.

12. NCCN 2026 melanoma context — neoadjuvant contrast

NCCN Melanoma: Cutaneous v3.2026 elevates neoadjuvant systemic therapy as preferred initial approach for resectable nodal disease — pembrolizumab or ipilimumab+nivolumab (Cat 1), with other IO options and BRAF/MEKi in certain BRAF V600 circumstances. This cutaneous paradigm should be cited as contrast: mucosal primaries are not the evidence base for S1801/NADINA; resectability and local control constraints differ.

Still use NCCN cutaneous systemic agents as the practical drug menu for MM while stating evidence quality is extrapolated. Mucosal-specific guideline sections and European site-specific guidance may supplement.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Melanoma: Cutaneous v3.2026; Head and Neck Cancers v2.2026 (sinonasal context)
- Cutaneous resectable nodal: neoadjuvant CPI preferred (pembro; ipi+nivo Cat 1) — do not blindly equate to mucosal primaries.
- Metastatic melanoma systemic options (PD-1 ± CTLA-4/LAG-3; BRAF/MEKi if V600) used for MM by extrapolation.
- Emphasize molecular testing (BRAF/KIT/NRAS) in non-cutaneous subtypes.
- Local therapy remains decisive for mucosal sites despite systemic advances.

13. irAE & supportive care in MM patients

Standard irAE algorithms apply (skin, colitis, hepatitis, endocrinopathy, pneumonitis). Head-and-neck RT fields plus CPI may intensify mucositis — coordinate supportive care. Post-surgical wounds need infection surveillance during steroids for irAEs.

- Vaccinate appropriately before IO when feasible.
- Thromboprophylaxis individualized for pelvic surgery patients.

14. Summary checklist & exam pearls

End with a crisp comparison table: cutaneous vs mucosal across UV, TMB, BRAF, KIT, CPI ORR, neoadjuvant evidence, prognosis.

- Sites: H&N;, anorectal, vulvovaginal
- Molecular: KIT/NRAS > BRAF
- Local: surgery ± RT; margins often limited
- Systemic: CPI first; imatinib if KIT; BRAF/MEKi if V600
- Prognosis worse than cutaneous
- NCCN 2026 neoadjuvant CPI = cutaneous nodal story — contrast explicitly

MM 30-second revision flow

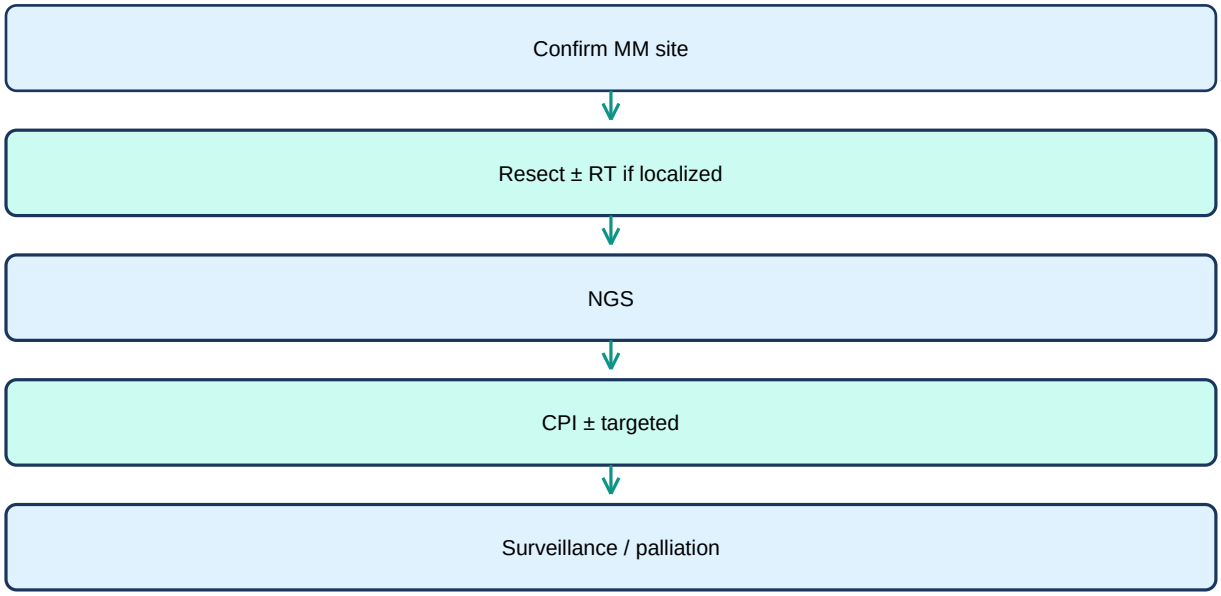


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Uveal melanoma is a different disease — do not conflate.
- Quote CheckMate/KEYNOTE mucosal subgroup humility (lower ORR).
- KIT testing is the targeted-therapy doorway.
- Draw site map + molecular pie chart for quick marks.
- State MDT site specialty by anatomic primary.

— Additional exam-length revision modules —

The following modules reinforce algorithms, toxicities, Indian/NCG context, and last-page summary structure.

3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.
- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).
- Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.

Work-up algorithm (draw this in the answer booklet)

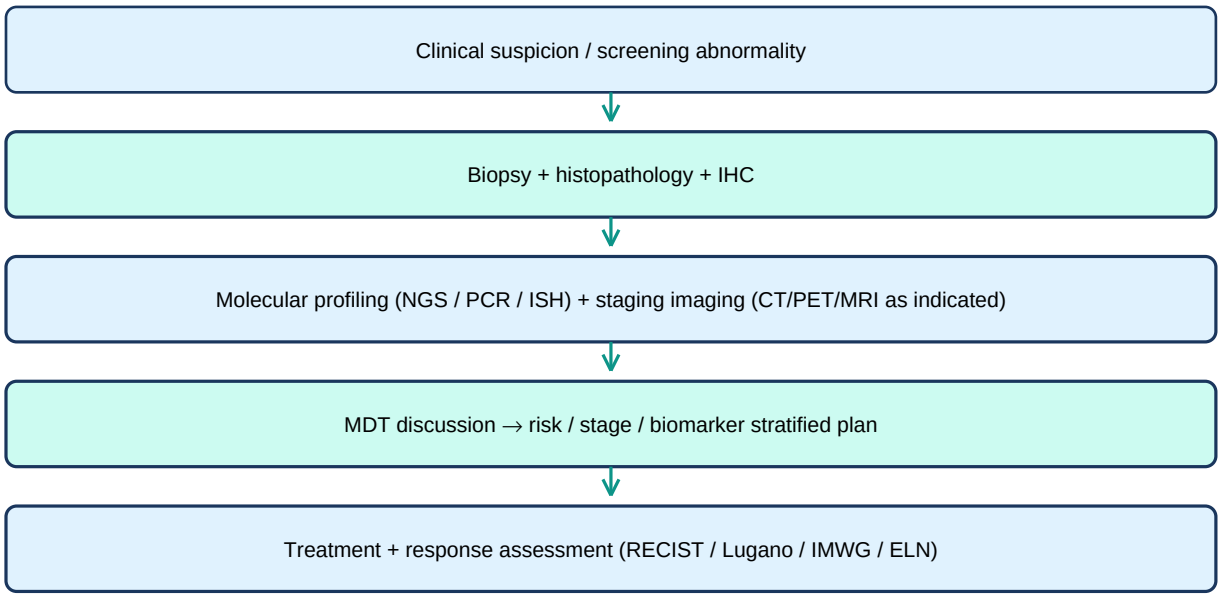


Figure: stepwise exam algorithm

4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

- Localized disease: multimodality — surgery/RT/systemic as per disease chapter.
- Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).
- Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.
- Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).
- Re-biopsy / liquid biopsy at progression to map resistance.

Management flowchart (reproduce in exam)

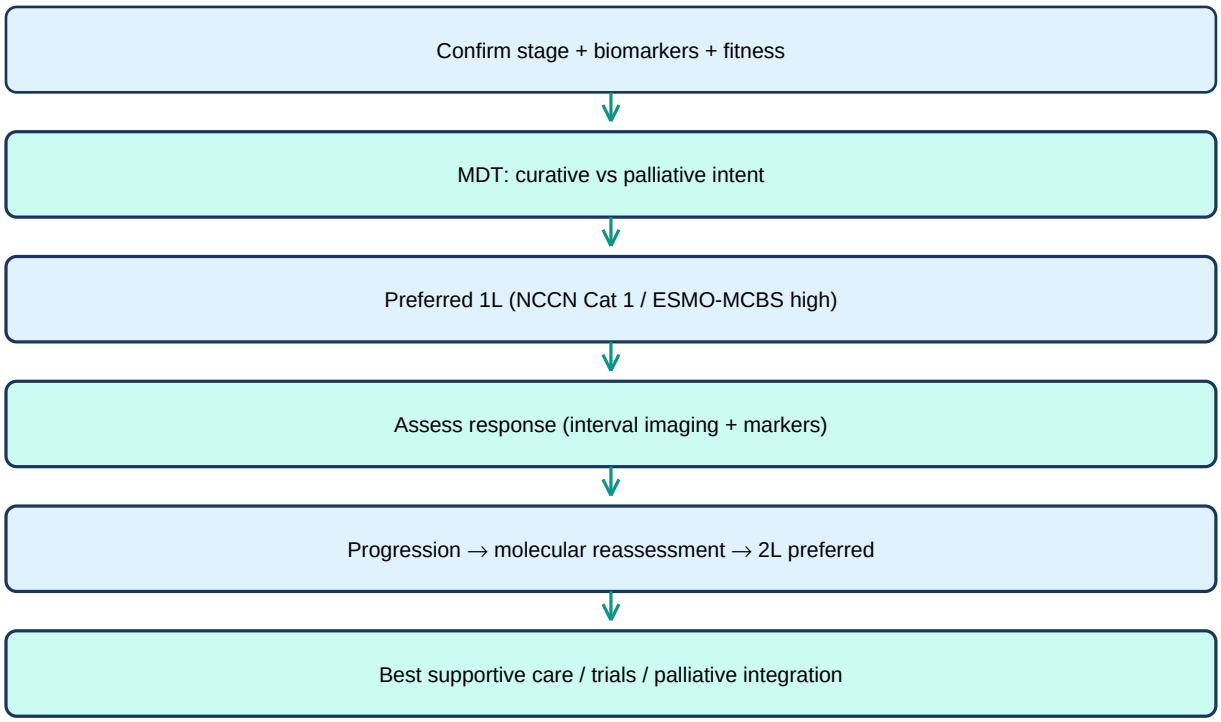


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

- Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.
- Distinguish statistical vs clinical significance; note crossover and subgroup caveats.
- For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

Trial	Key result / exam takeaway
CheckMate 067	Nivo + ipi vs nivo vs ipi
KEYNOTE-006	Pembrolizumab vs ipi
RELATIVITY-047	Nivo + relatlimab
COMBI-d/v / COLUMBUS	BRAF/MEK inhibitors
SWOG S1801 / NADINA	Neoadjuvant CPI → preferred resectable nodal NCCN 2026

Table: landmark clinical trials

6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

• Anticipate class toxicities and counsel patients; use CTCAE grading.

• Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.

• IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.

• ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.

• TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.

• Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.

Toxicity management loop

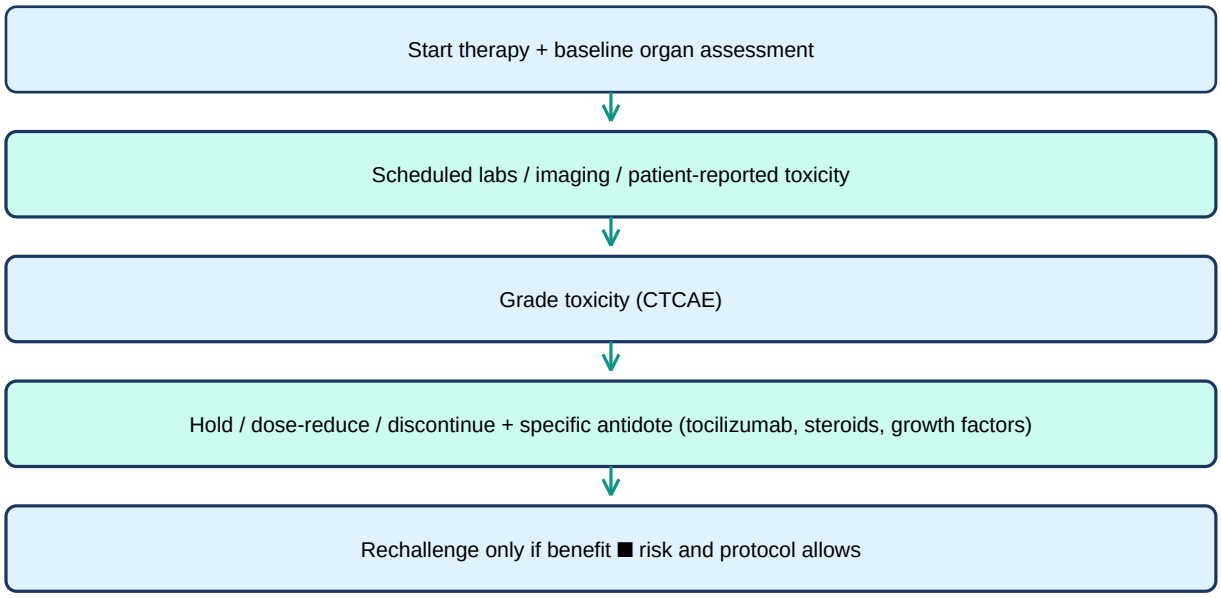


Figure: stepwise exam algorithm

7. NCCN 2026 / 2027 updates (high-yield for ‘recent advances’)

Paper IV rewards currency. Explicitly write: **“As per NCCN Guidelines updates in 2026/2027…”** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

• State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.

• If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* Melanoma: Cutaneous v3.2026

• Neoadjuvant systemic therapy now preferred initial approach for resectable nodal disease (vs surgery-first historically).

• Preferred neoadjuvant: pembrolizumab; ipilimumab + nivolumab (Cat 1). Other: nivo alone; nivo + relatlimab. BRAF/MEKi useful in certain circumstances for BRAF V600.

• Earlier-line CPI expansion: ATOMIC stage III dMMR colon; neoadjuvant CPI for dMMR rectal; neoadjuvant melanoma; EV+P bladder; FLOT+durvalumab gastric.

• PD-L1 testing emphasized for gastric/EGJ; limited benefit if PD-L1 negative.

EXAM PEARLS / HALLMARKS

• NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).

• Category 1 = high-level evidence + uniform NCCN consensus.

8. Special situations, differentials & practical issues

Discuss differentials, mimics, and when to refer to radiation oncology, surgery, palliative care, genetics, or cell-therapy centers.

Include brief notes on health economics (cost-effectiveness, biosimilars) and clinical trial enrollment as standard options.

- MDT documentation improves outcomes and exam completeness.
- Survivorship care plans: late effects, second malignancies, cardiac, endocrine, psychosocial.
- Real-world evidence and digital tools increasingly complement RCT data (relevant to 2026 Paper IV themes).

9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

- Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.
- Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.
- Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.
- Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

EXAM PEARLS / HALLMARKS

- Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)
- Mention number-needed-to-treat conceptually if space permits.

10. Summary checklist (write this last page in the booklet)

Close the 15-mark answer with a crisp numbered summary the examiner can tick.

- Definition + epidemiology one-liner
- Key biology / hallmark diagram
- Essential diagnostics & biomarkers
- Risk/stage stratification
- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

30-second revision flowchart

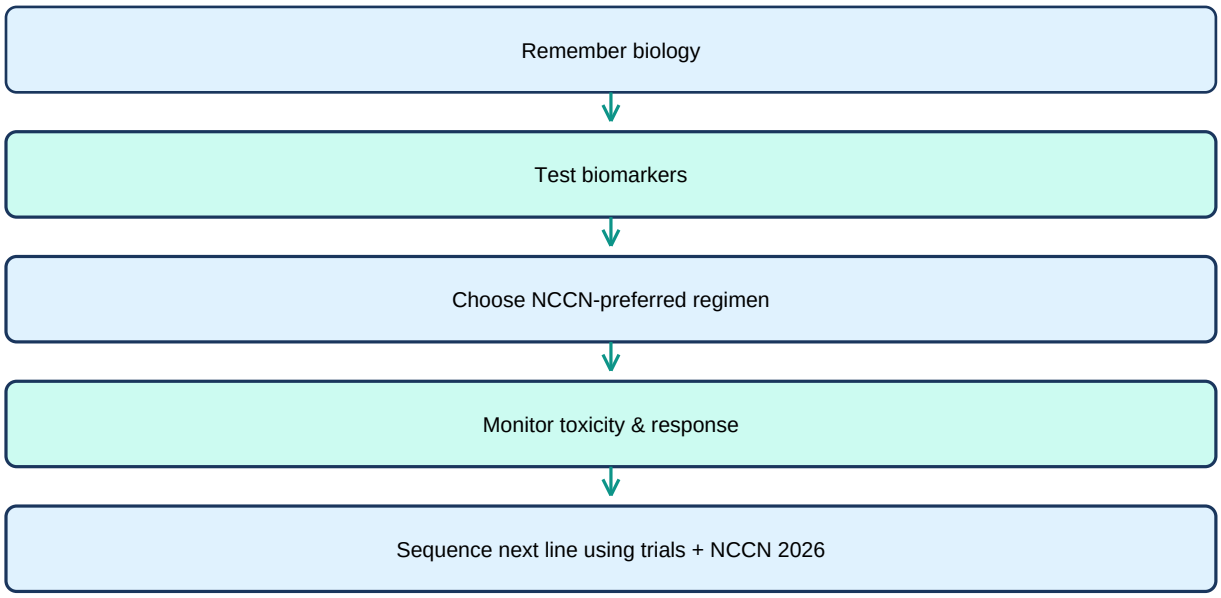


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say 'NCCN 2026' explicitly for Paper IV.

11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.
- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.
- PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.

• ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

Clinical algorithm

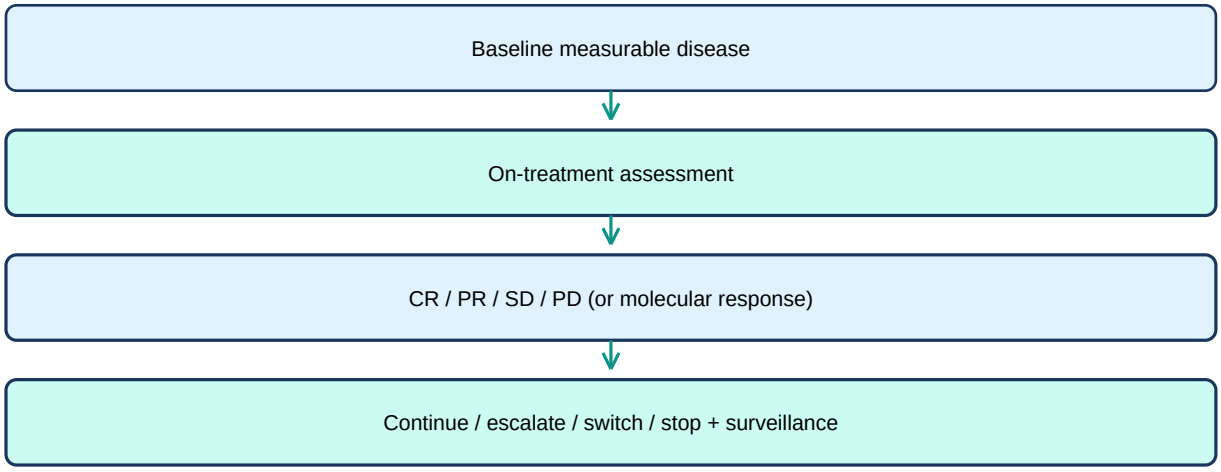


Figure: stepwise exam algorithm

13. Comparison tables useful for exams

Draw 2×N tables comparing old standard vs new standard; or drug A vs B (efficacy, toxicity, CNS activity, cost/access).

• Column ideas: target, dose, PFS, OS, grade ≥3 AEs, CNS activity, NCCN category.

• Another table: resistance mutation → preferred next drug.

Trial	Key result / exam takeaway
CheckMate 067	Nivo + ipi vs nivo vs ipi
KEYNOTE-006	Pembrolizumab vs ipi
RELATIVITY-047	Nivo + relatlimab
COMBI-d/v / COLUMBUS	BRAF/MEK inhibitors
SWOG S1801 / NADINA	Neoadjuvant CPI → preferred resectable nodal NCCN 2026

Table: landmark clinical trials

14. Indian context, NCG & practical exam tips

National Cancer Grid (NCG) resource-stratified guidelines are frequently expected in Indian DM exams. Discuss biosimilars (trastuzumab, rituximab), generic TKIs, and center-based cell therapy availability.

• Document MDT decisions; clinical trial referral when standard options exhausted.

• Counsel financial toxicity transparently; prioritize highest-value Category 1 options.

• Infection prophylaxis algorithms crucial in hematologic topics (antifungal, antiviral, PJP).

15. Sample ‘examiner hot spots’ & model closing paragraph

Model closing: “In summary, management of this problem is biomarker-driven. Current standards are supported by phase III trials and reflected in NCCN 2026/2027 updates. Sequencing should maximize OS and QoL while mitigating class-specific toxicities. Enrollment in clinical trials and adherence to MDT pathways remain integral.”

Hot spots: exact doses, one HR/median, resistance mechanism, NCCN category, and a drawn algorithm.

EXAM PEARLS / HALLMARKS

- If stuck on memory: write mechanism → class → one trial → one dose → one toxicity → NCCN 2026 line.
- Never contradict yourself between algorithm and trial table.

A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

Therapeutic eras timeline

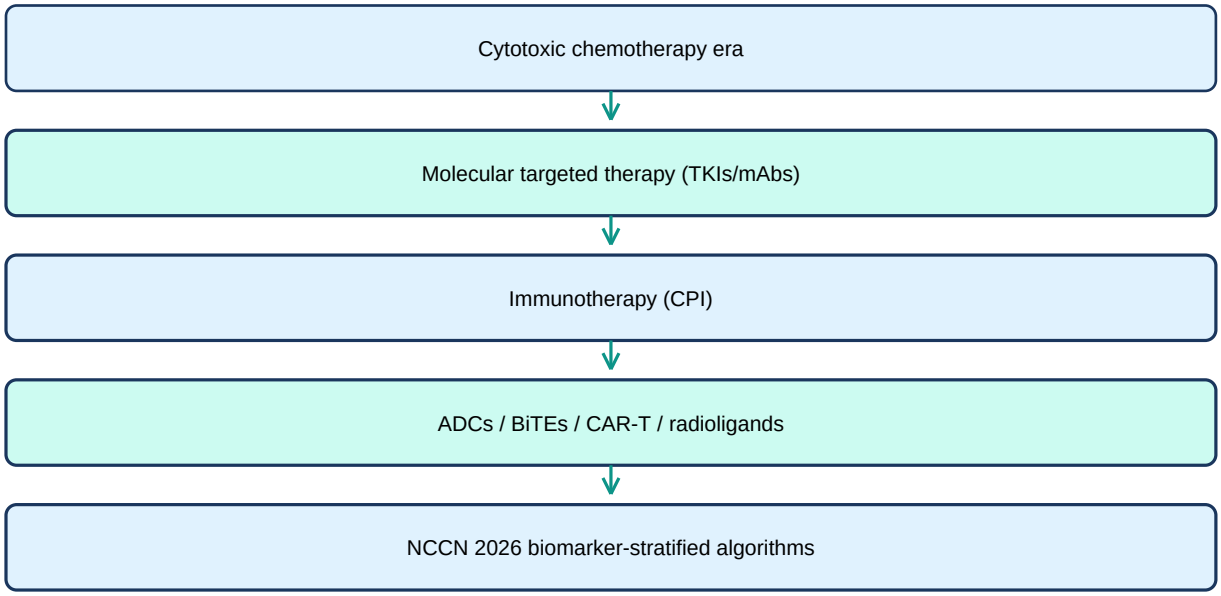


Figure: stepwise exam algorithm

C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

Morphology	Light microscopy pattern
IHC panel	Lineage + predictive markers
Cytogenetics	Karyotype / FISH fusions
NGS	SNVs, indels, CNAs, MSI, TMB
Integrated Dx	WHO/ICC + AJCC stage

Figure: mechanism / pathway map

D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

Clinical algorithm

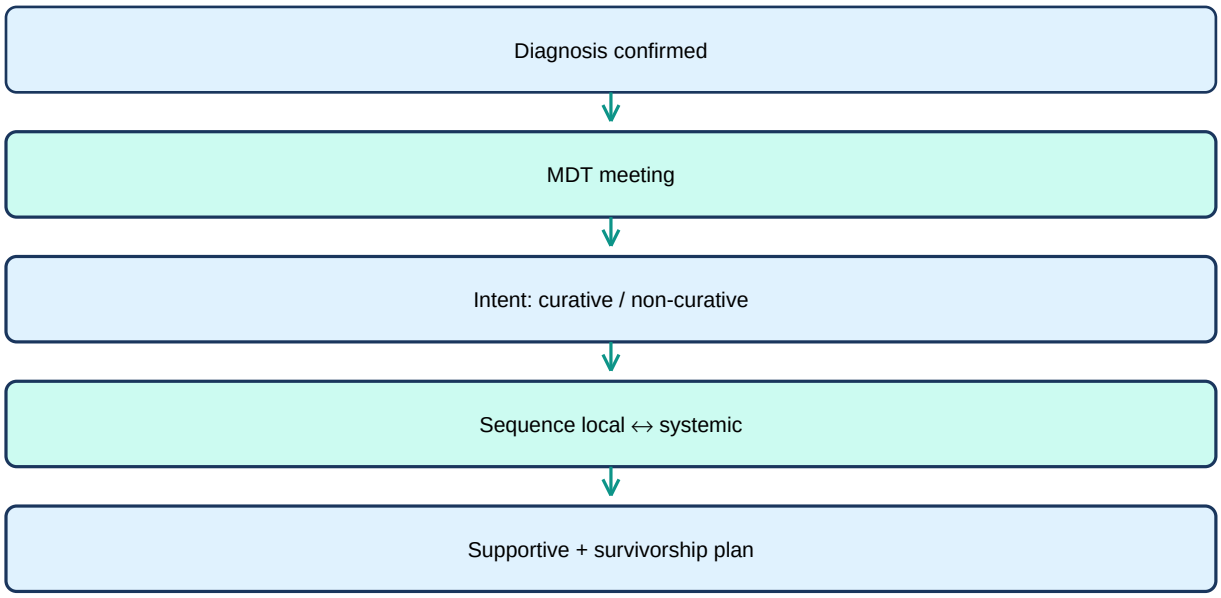


Figure: stepwise exam algorithm

E. Immunotherapy deep notes

CTLA-4 acts priming phase in lymphoid organs; PD-1/PD-L1 in peripheral effector phase. LAG-3 (relatlimab) additional checkpoint.

Biomarkers: PD-L1, MSI/dMMR, TMB — imperfect. TME, gut microbiome modulation is an exam discussion point (2026 short note theme).

Neoadjuvant IO preferred for resectable nodal melanoma (NCCN 2026). Combinations with VEGF in RCC; with chemo in lung/gastric; with ADC in bladder (EV+P).

- irAE timelines: skin/GI early; endocrine anytime; late neuropathies/myocarditis rare but deadly.
- Rechallenge decisions individualized after grade 2 resolution; avoid after grade 4 myocarditis/SJS etc.

G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.

- Underline drug doses.
- Box the NCCN 2026 bullets.

EXAM PEARLS / HALLMARKS

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.
- NCCN 2026 phrase is expected in Paper IV recent advances.

K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).
- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

Clinical algorithm

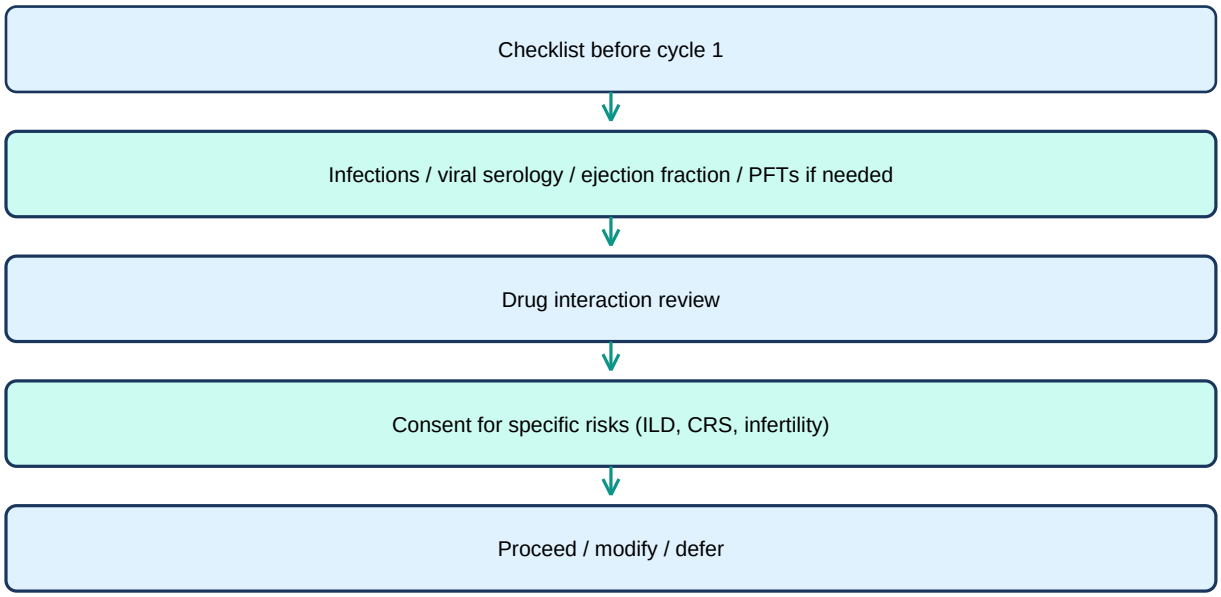


Figure: stepwise exam algorithm

L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for

resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.
- Vignette 3: progression with resistance mutation — sequenced next line.
- Vignette 4: severe toxicity — hold, treat, rechallenge decision.

M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

Quadrant 1	Hallmark gene / pathway / figure
Quadrant 2	Must-order tests + staging
Quadrant 3	1L / 2L / 3L with doses
Quadrant 4	Trials + NCCN 2026 + toxicities
Center	MDT + patient goals

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

- Wall chart + flowchart + trial table = visual completeness.
- Write ‘NCCN 2026’ in quadrant 4 explicitly.

I2. Full-length essay addendum — tables to copy into the answer book

For 15-mark elaborations, add two hand-drawn tables: (Table A) Drug–dose–pearl; (Table B) Trial–population–endpoint–practice change. Then write a short paragraph on quality-of-life and patient-reported outcomes, which modern trials increasingly report.

Discuss biomarker evolution on therapy (liquid biopsy cadence), and when to stop therapy (duration of CPI, TFR in CML, maintenance stopping rules).

Include a brief note on second malignancies and late effects when curative-intent therapy is discussed (alkylators, RT fields, PARPi MDS/AML risk, CHL survivors).

- Table A columns: Drug | Dose | Key toxicity | Pearl
- Table B columns: Trial | Arm | Primary endpoint | Result | Change in practice
- One paragraph on shared decision-making and financial counseling

Trial	Key result / exam takeaway
CheckMate 067	Nivo + ipi vs nivo vs ipi
KEYNOTE-006	Pembrolizumab vs ipi
RELATIVITY-047	Nivo + relatlimab
COMBI-d/v / COLUMBUS	BRAF/MEK inhibitors
SWOG S1801 / NADINA	Neoadjuvant CPI → preferred resectable nodal NCCN 2026

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* Melanoma: Cutaneous v3.2026

• Neoadjuvant systemic therapy now preferred initial approach for resectable nodal disease (vs surgery-first historically).

• Preferred neoadjuvant: pembrolizumab; ipilimumab + nivolumab (Cat 1). Other: nivo alone; nivo + relatlimab. BRAF/MEKi useful in certain circumstances for BRAF V600.

• Earlier-line CPI expansion: ATOMIC stage III dMMR colon; neoadjuvant CPI for dMMR rectal; neoadjuvant melanoma; EV+P bladder; FLOT+durvalumab gastric.

• PD-L1 testing emphasized for gastric/EGJ; limited benefit if PD-L1 negative.

I. Extended notes — resistance, sequencing & future directions

Resistance biology should occupy a dedicated half-page: on-target secondary mutations, bypass signaling, lineage plasticity (e.g., EGFR NSCLC → SCLC), antigen loss after cellular therapy, and immunosuppressive TME changes after CPI.

Sequencing principles: use the agent with largest OS benefit early when toxicities allow; reserve cross-resistant drugs; consider cumulative neuropathy/cardiotoxicity; plan CNS-active agents if brain metastases are present or likely.

Future directions appropriate for 2026 answers: tumor-agnostic indications (NTRK, RET, BRAF, MSI-H, TMB-H, HER2-mut), radioligand therapy, CAR-NK/off-the-shelf products, ctDNA-guided escalation/de-escalation, AI decision-support, and microbiome modulation adjuncts.

Health services angle: real-world evidence validating trial results in older/comorbid patients; digital remote toxicity monitoring; and equity of molecular testing access — themes also reflected in recent Paper IV questions.

- Write one ‘if progression then…’ resistance table.
- Mention at least one tumor-agnostic approval if biology overlaps.
- Close with trial enrollment as a standard option.

J. Quick-revision flash cards (last 1–2 pages)

Convert the topic into 8–10 flash cards mentally: (1) definition (2) hallmark gene (3) key test (4) 1L preferred (5) dose (6) trial (7) top toxicity (8) NCCN 2026 update (9) Indian/NCG pearl (10) one differential.

- Flash 1: One-line definition
- Flash 2: Driver / pathway
- Flash 3: Mandatory biomarker
- Flash 4: NCCN preferred 1L
- Flash 5: Exact dose
- Flash 6: Landmark phase III + endpoint
- Flash 7: Grade ≥3 toxicity to warn
- Flash 8: 2026 guideline change

- Flash 9: Resistance next step
- Flash 10: Supportive care must-do

EXAM PEARLS / HALLMARKS

- If time-poor in the hall: draw algorithm + trial table + NCCN box — captures most marks.
- Keep handwriting large for doses and HRs.

Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

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Final NCCN 2026 box

Copy these bullets onto the last page of your answer booklet for Paper IV currency marks.

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Disclaimer: Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.